Session 4 · Open-book AI (RAG): demo runbook

Saturday 7 Nov 2026 · Applied AI & LLM Engineering · Everything runs in a browser, no code.

Before class (10 min):

1. Open **RAG Lab** on the projector laptop. Click **Ask** once and back to **Library**. It remembers settings in the browser, so reset by re-selecting: Open book · 3 pieces · Small · overlap off · latest off · "answer only from pieces" on.
2. Create a NotebookLM notebook called **Lakeview Helpdesk** and upload all 8 PDFs from the `documents` folder. Wait until all 8 show as sources.
3. Open a second tab with ChatGPT, Claude or Gemini, **with no files attached**.
4. Put the 8 PDFs somewhere students can download them (shared Drive folder or WhatsApp group).
5. Check that notebooklm.google.com loads on the classroom Wi-Fi. If it doesn't, RAG Lab covers every demo offline.

Demo 1 · Closed book vs open book ~6 min · after slide "Same question, two ways"

1. In the plain chatbot tab, ask: `At Lakeview Institute of Business Studies, what is the fine for returning a library book late?`
   Expect: a guess ("usually ₹1–2 per day") or "I don't have information about this college". *Ask the room: "Can we trust this? Where did it come from?"*
2. In NotebookLM, ask the same question.
   Expect: ₹5 per day, with a small citation number. **Hover or click the number**: it shows the exact passage from Library Rules.
3. *"Same AI ability. The difference is that we handed it the right page first. That's RAG."*

Backup (no internet): RAG Lab → Ask → Q1. Switch **Closed book** (made-up answer) then **Open book** (₹5, cited). Click **Check this answer** each time.

Demo 2 · Look inside the pipeline ~8 min · after slide "Two stages"

1. **Library tab.** *"This is everything our assistant can read."* Point at the two **Attendance Policy** cards: June 2023 and July 2026. Don't explain yet; it comes back in Demo 3.
2. **Cut into pieces tab** → Library Rules, Small. Show the ✂ marks. *"The computer cuts every 30 words. It has no idea where a rule ends."* Point at pieces 3 and 4: the laptop rule is cut in half. Leave it there.
3. **Ask tab → Q6** (fees back if I drop out). Walk the three stages:
   - **1 · Find:** the top piece says "withdraws within the first 15 days", yet the student said "drop out" and "second week". *"Meaning search from last week at work."*
   - **2 · Pack:** open *Show the full message sent to the AI*. Read the instruction aloud: *use ONLY the pieces, cite them, say you don't know*.
   - **3 · Answer:** 75% back, with [2]. Click **[2]**: it jumps to the piece. Then **Check this answer**.

Demo 3 · Break it on purpose ~10 min · after slide "Where it breaks"

Before each step, ask the room to **predict** whether the answer will be right. Then click **Check this answer**.

Failure	Clicks in RAG Lab (Ask tab)	What to say
A. Rule cut in half	Q2 (laptop for 3 days in exam week) at default settings → **"Yes"** with a citation. Check → **Wrong**. Turn on **Overlap** → "No… same evening by 8 pm" → Correct. (Or set piece size to Medium.)	"It has a citation and it's still wrong. A citation proves where a sentence came from, not that the answer is complete."
B. Old document	Q3 (minimum attendance) → reply leads with **75%** (the 2023 rule). Check → Careful. Set Pieces to fetch to **1** → flat 75%, Wrong. Tick **Use only the latest version** → 80%, Correct. Set Pieces back to **3**.	"The old file was worded more like the question, so it ranked first. Old documents left in the folder are a real problem in companies."
C. Answer not in the documents	Q4 (hostel Wi-Fi password) → "The documents don't give a password… IT help desk" → Correct. Untick **Answer only from the pieces** → invents **Lakeview@2026** → Wrong.	"One line of instruction decides whether it admits it doesn't know or makes something up."
D. Too little fetched	Q6, set Pieces to fetch to **1** → "I couldn't find this" → Careful.	"Honest, but useless. The answer existed; we just didn't fetch enough."

Live version in NotebookLM (optional, 2 min): ask `What is the hostel Wi-Fi password?` (should say it's not in the sources) and `What minimum attendance do I need to sit for exams?`. It sees both attendance PDFs, so watch whether it mentions 75%, 80% or both. Whatever it does, it's a discussion point.

Student exercise · Be the fact-checker ~12 min · groups of 3

1. Open NotebookLM (free with a Google account). Any chatbot that accepts file uploads also works.
2. Upload **three** of the Lakeview PDFs. Each group picks a different three.
3. Ask three questions and fill the table: one the documents **can** answer, one they **can't**, and one **tricky** question in your own words (different words from the document).
4. For every answer, open the citation and decide: does the cited text really support the answer?

Type	Our question	AI's answer (short)	Cited?	Does the citation support it? Yes / Partly / No
Can answer				
Can't answer				
Tricky wording				

Question ideas if a group gets stuck

Document	Can answer	Can't answer	Tricky wording
Placement Cell	What score do I need to register for placements?	Which companies are visiting this year?	I already have a job offer of ₹6 lakh. Can I still sit for a ₹8 lakh company?
Clubs and Events	When is the annual fest?	Who is the president of the Finance Club?	Our club wants to call a guest speaker on 10 March. When must we apply?
Hostel Handbook	What time do hostel gates close?	How much is the hostel fee?	Can my parents stay over in my room this weekend?
Examinations	When are hall tickets released?	When do end-term exams start?	I flunked one paper. When can I write it again and what will it cost?
Library Rules	How many books can I borrow?	Does the library have the Harvard Business Review?	I'm going home for a week. Can I keep a laptop over the break?
Fee Refund	How do I apply for a refund?	When is the next fee payment due?	I'm leaving the hostel. When do I get my deposit back?

Debrief (3 min): ask two groups for a "No" in the last column. Every one of those is a lesson: wrong piece fetched, piece cut off, or the AI added something.

Answers to two tricky ones: *₹8 lakh is less than 1.5 × ₹6 lakh (= ₹9 lakh), so no. 10 March needs a proposal by 17 February (3 weeks before), plus Dean approval for an outside speaker.*

Exit question last 3 min

"Our assistant gave a wrong answer. Name three different places in the pipeline where it could have gone wrong."

Looking for: the right document wasn't in the library (or an old one was) · the search fetched the wrong pieces, or a rule was cut in half · the AI misread the pieces or added its own guess.

Next week: how to give our assistant a report card, i.e. measuring whether a change made it better or worse.